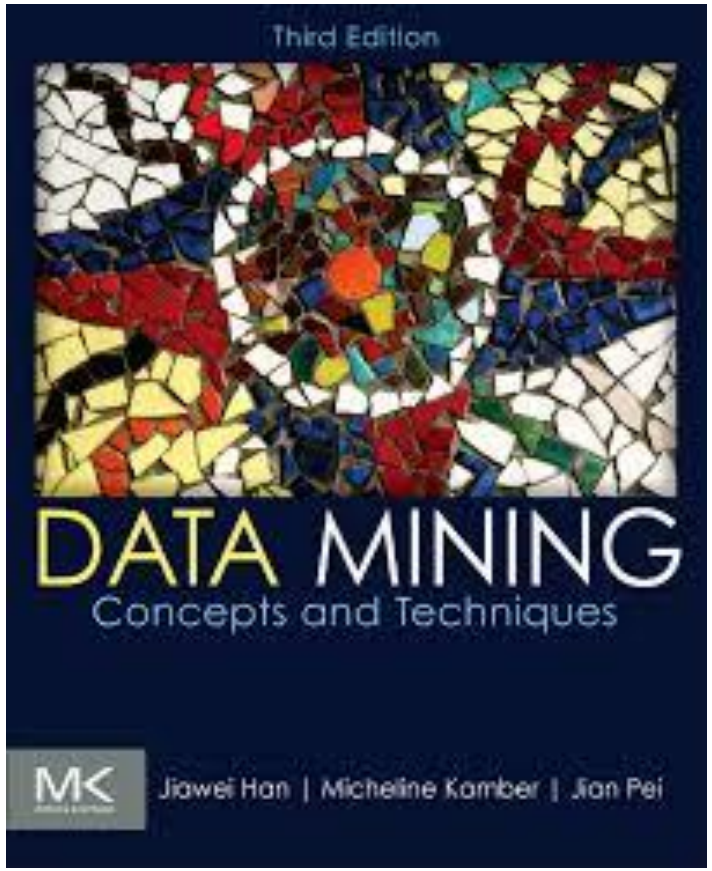
Machine Learning and Data Mining Basics

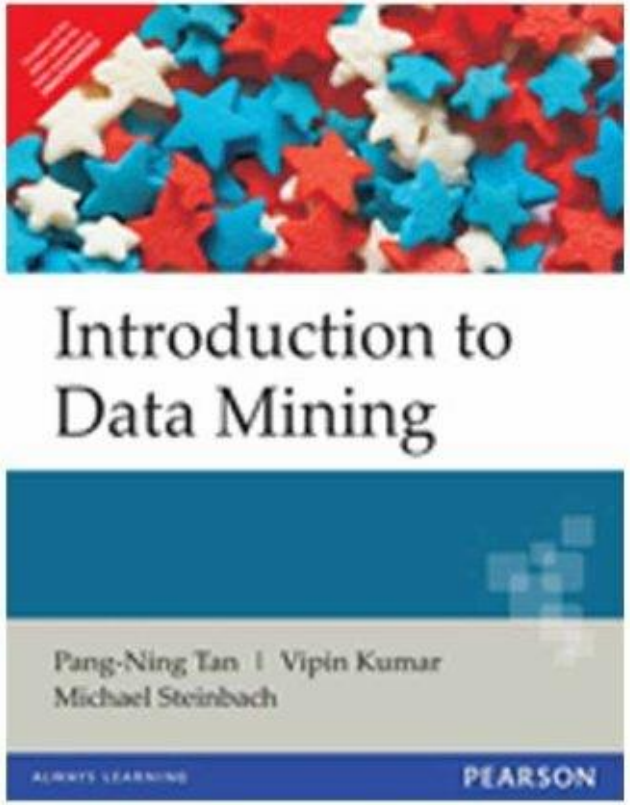
Course Goal

- **Differentiate Data Mining and Machine Learning** – Understand the scope, purpose, and key differences.
- **Understand Data Representation** – Learn how data is structured, formatted, and encoded for ML.
- **Master Data Preprocessing** – Handle missing data, scaling, normalization, and feature engineering.
- **Comprehend the Machine Learning Life Cycle** – From problem definition to model deployment.
- **Identify Types of Machine Learning** – Supervised, Semi-supervised, Unsupervised, and Reinforcement Learning.
- **Implement Core ML Algorithms** – Linear & logistic regression, decision trees, SVM, KNN, Naïve Bayes, etc.
- **Evaluate ML Models** – Metrics, validation, cross-validation, and performance improvement techniques.
- **Introduction to Deep Learning** – Basics of neural networks and their applications.
- **Develop Practical Thinking Skills** – Learn frameworks for problem-solving and common beginner pitfalls.

Course Materials and References

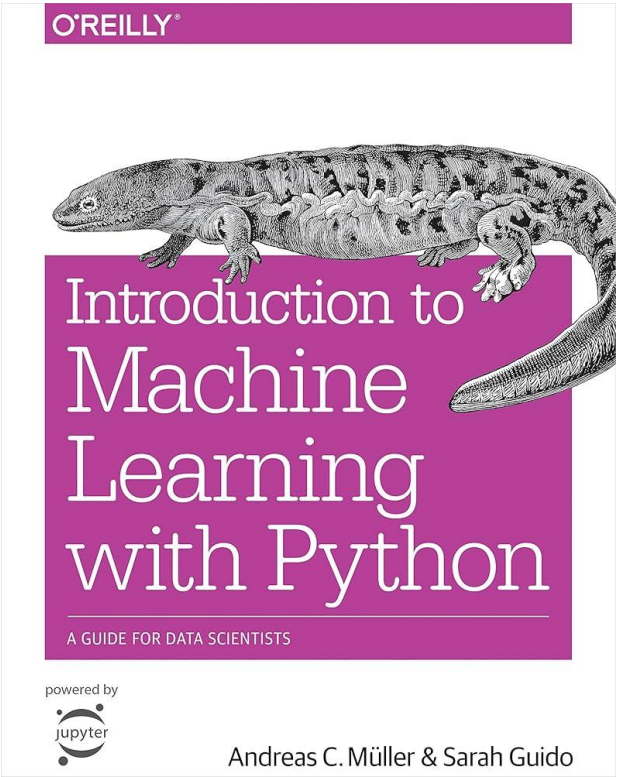


Data Mining Concepts and Techniques by Jiawei Han, Micheline Kamber, Jian Pei ·

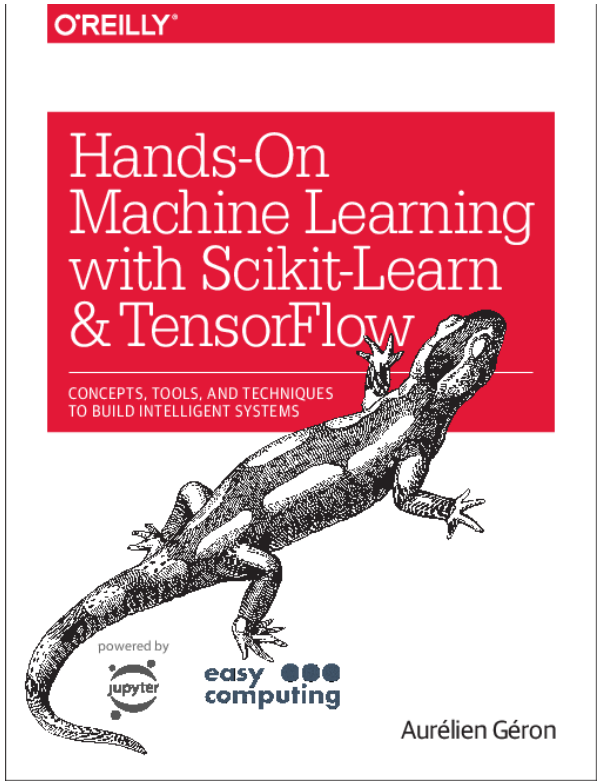


Introduction to Data Mining by Pang-Ning Tan Michael Steinbach Vipin Kumar

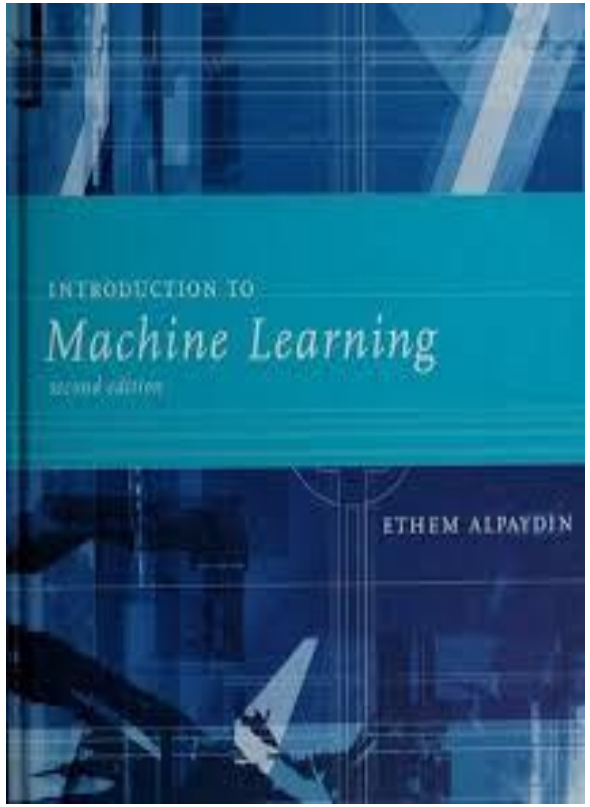
Course Materials and References



Introduction to Machine Learning with Python by Andreas C. Muller and Sarah Guido



Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurélien Géron



Introduction to Machine Learning by Ethem Alpaydın

Machine Learning

- What is Machine Learning
- Why Use Machine Learning
- Types of Machine Learning Systems
- Main Challenges of Machine Learning
- Machine Learning Cycle

- Classification
- Linear Models
- Underfitting vs Overfitting
- Model Evaluations
- K-Nnearest Neighbors Algorithm

What is Machine Learning

- Machine Learning is a subset of Data Mining focused on algorithms that learn patterns from data and generalize to unseen data.

- It gives computers the ability to learn without being explicitly programmed. (Arthur Samuel ,1959).

- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

What is Machine Learning

- ML focuses specifically on predictive modeling.

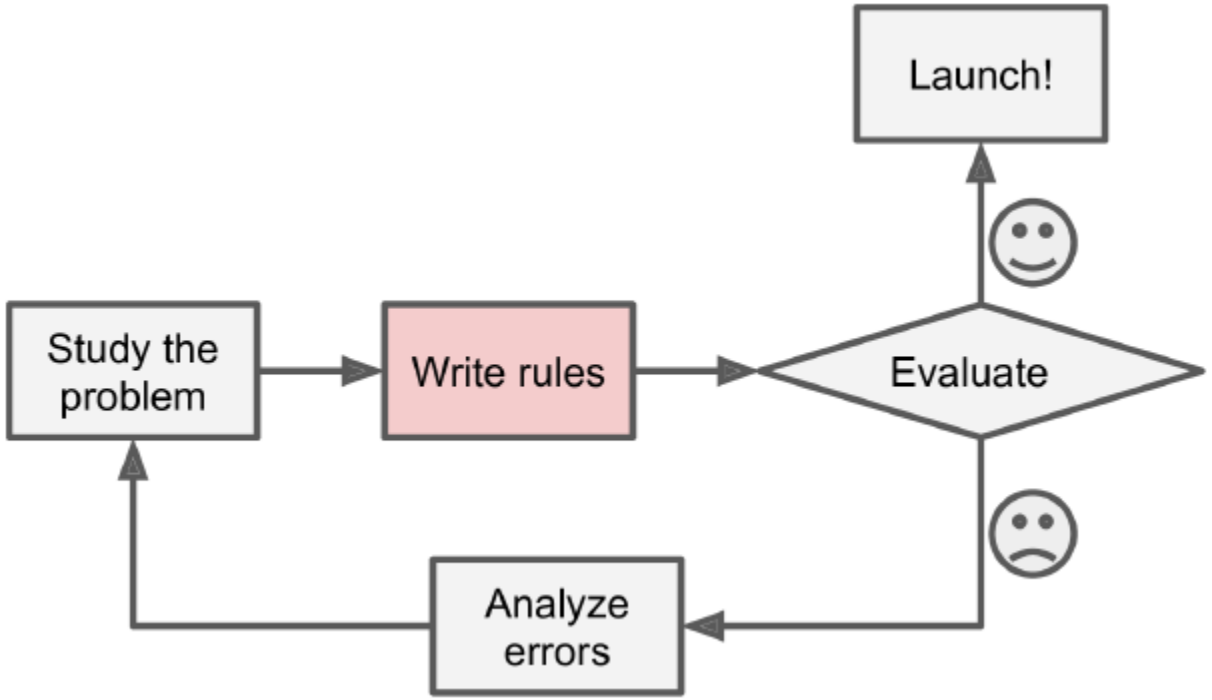
- Machine Learning:
  - Grew out of work in AI
  - New capability for computers
  - Data volume growth

Why Use Machine Learning?

- Take an example of spam filter

Traditional programming techniques:

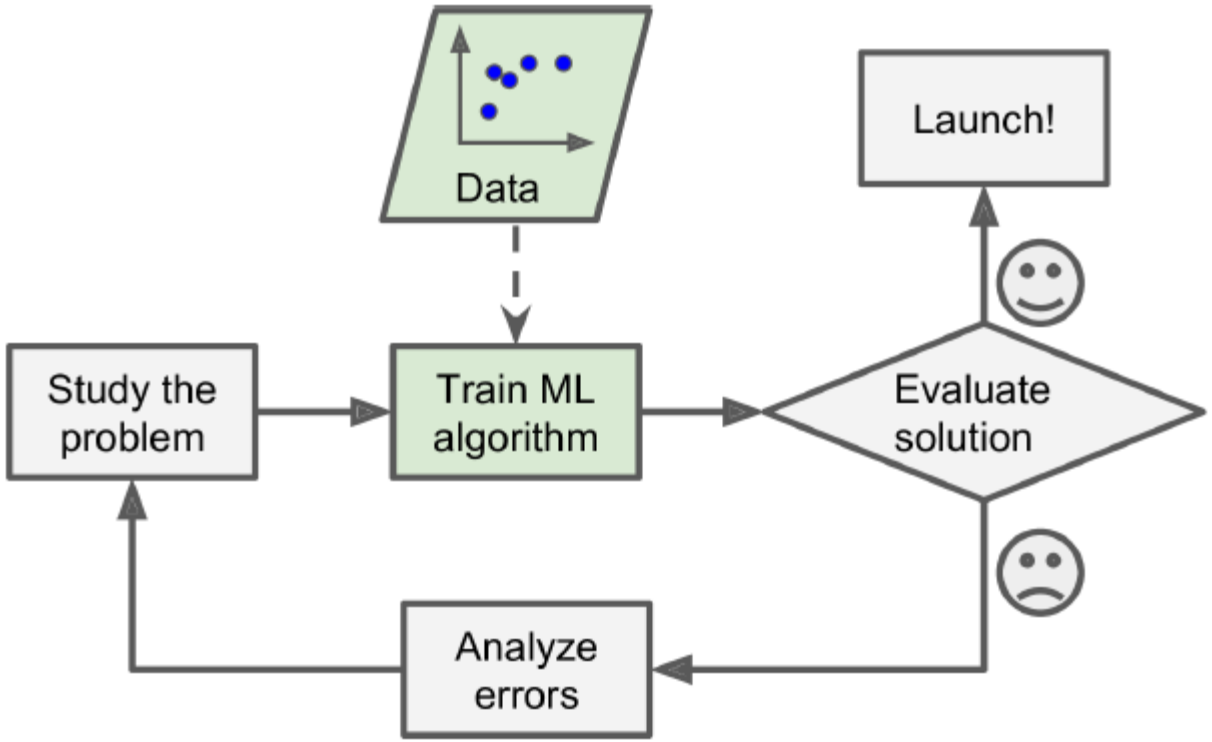
- Observe some words or phrases (“4U,” “credit card,” “free,” and “amazing”)
  – sender’s name, etc.
- Drawbacks:
  - Long list of complex rules
  - Hard to maintain



Why Use Machine Learning?

- ML Methods:
- Automatically learns which words and phrases are good predictors of spam
  - by detecting frequent patterns of words in the spam examples compared to the ham examples

- Advantages:
  - Shorter
  - Easier to maintain
  - More accurate



Machine Learning can be Used for:

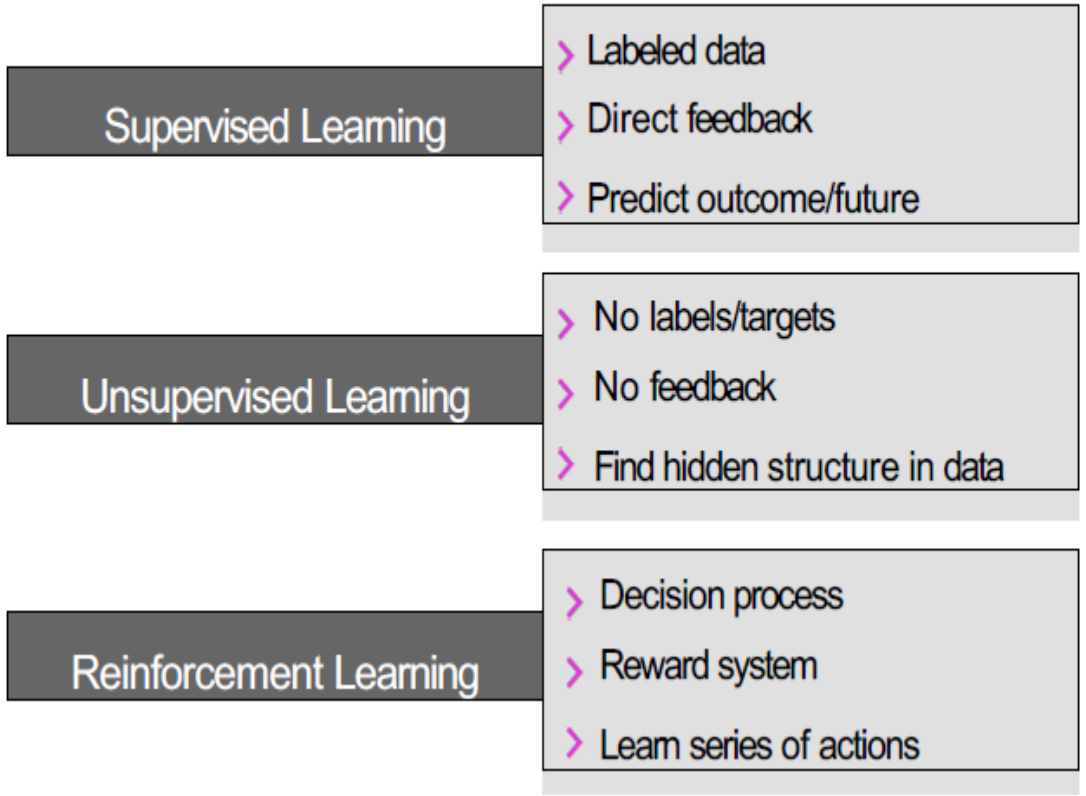
- Problems for which existing solutions require a lot of manual tuning or long lists of rules
  - Better performance
- Complex problems for which there is no good solution at all
- Fluctuating environments
  – Adaptive
- Getting insights about complex problems and large amounts of data
  – Discover unknown patterns

Types of Machine Learning Systems

- Trained with human supervision or not
  - supervised, unsupervised, semi-supervised, and Reinforcement Learning
- Can learn incrementally on the fly or not
  - Online (from a stream of incoming data) versus batch learning
- Working by comparing new data points to known data points, or detecting patterns in the training data and build a predictive model
  - instance-based versus model-based learning

Types of Machine Learning Tasks:

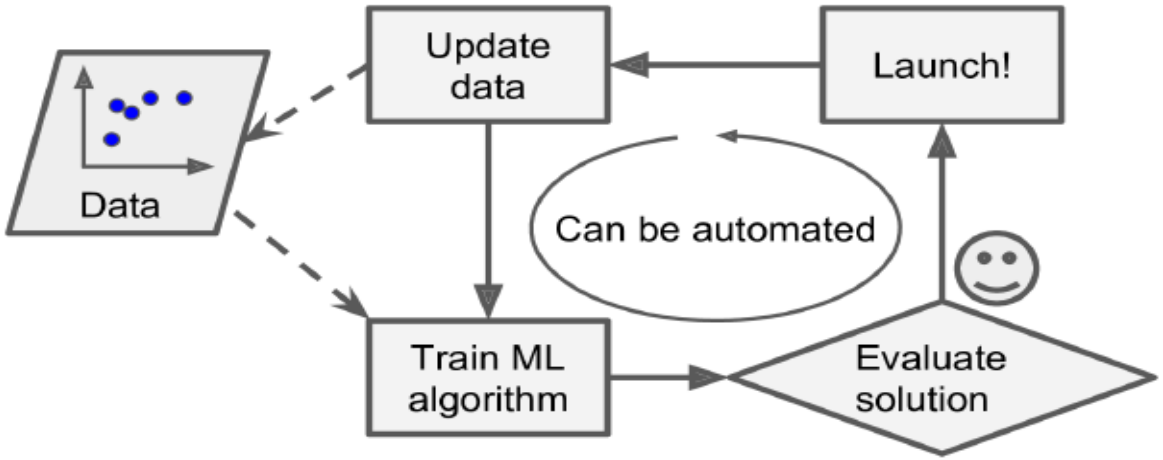
- ***Supervised Learning:***
  1. Regression (Continuous values)
  2. Classification (Categories)
- ***Unsupervised Learning:***
  1. Clustering
  2. Dimensionality reduction
  3. Anomaly detection
- Semi Supervised Learning
- Reinforcement Learning



Batch Learning

- It must be trained using all the available data (Offline learning).
- Properties
  - Usually takes a lot of time (many hours) and computing resources
  - Runs without learning anymore (after finishing training)
  - New data → Train a new version from the beginning on the whole dataset
  - The whole process can be automated
- Not suitable for:
  - Rapidly changing data
  - Huge amount of data
  - Limited resources

Solution: Online learning



Online Learning

- The system is trained incrementally by providing data sequentially
  - Raw by raw or in mini-batches
- Properties
  - Learning process is fast and cheap
  - Learning rate
    - High: rapidly adapt to new data
    - Low: less sensitive to noise

➢ Suitable for:

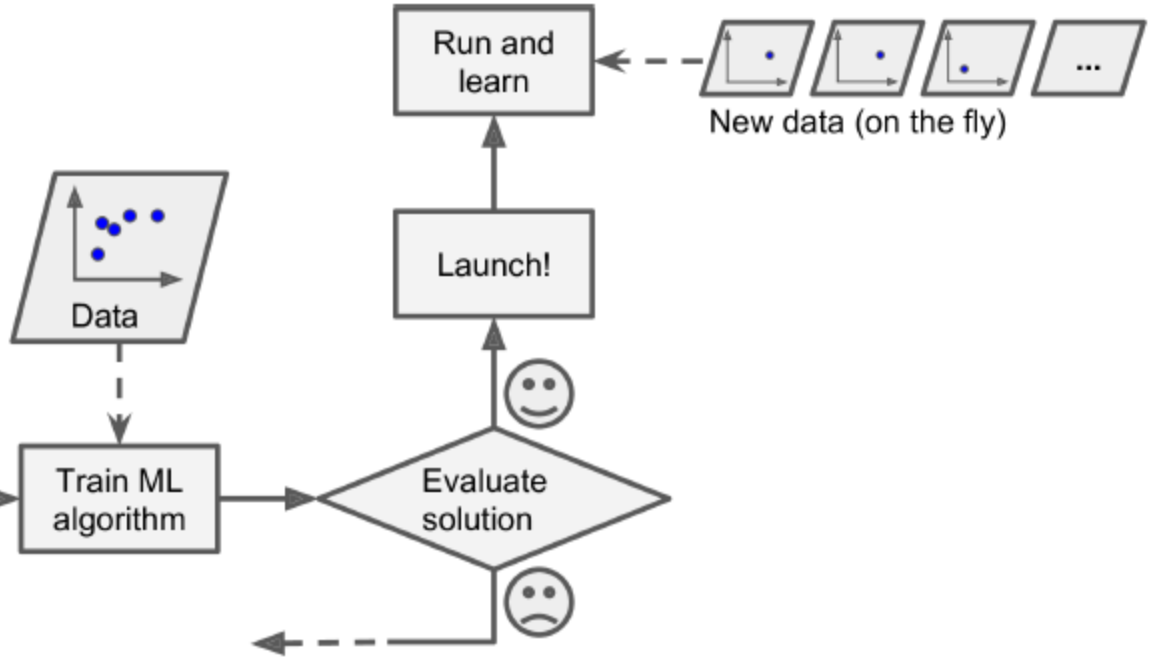
✓ Continuous data (e.g., stock prices)

✓ Rapidly changing data

✓ Autonomous learning

✓ Out-of-core learning (for huge data or limited resources)

  - Huge datasets that cannot fit machine's memory



Main Challenges of Machine Learning

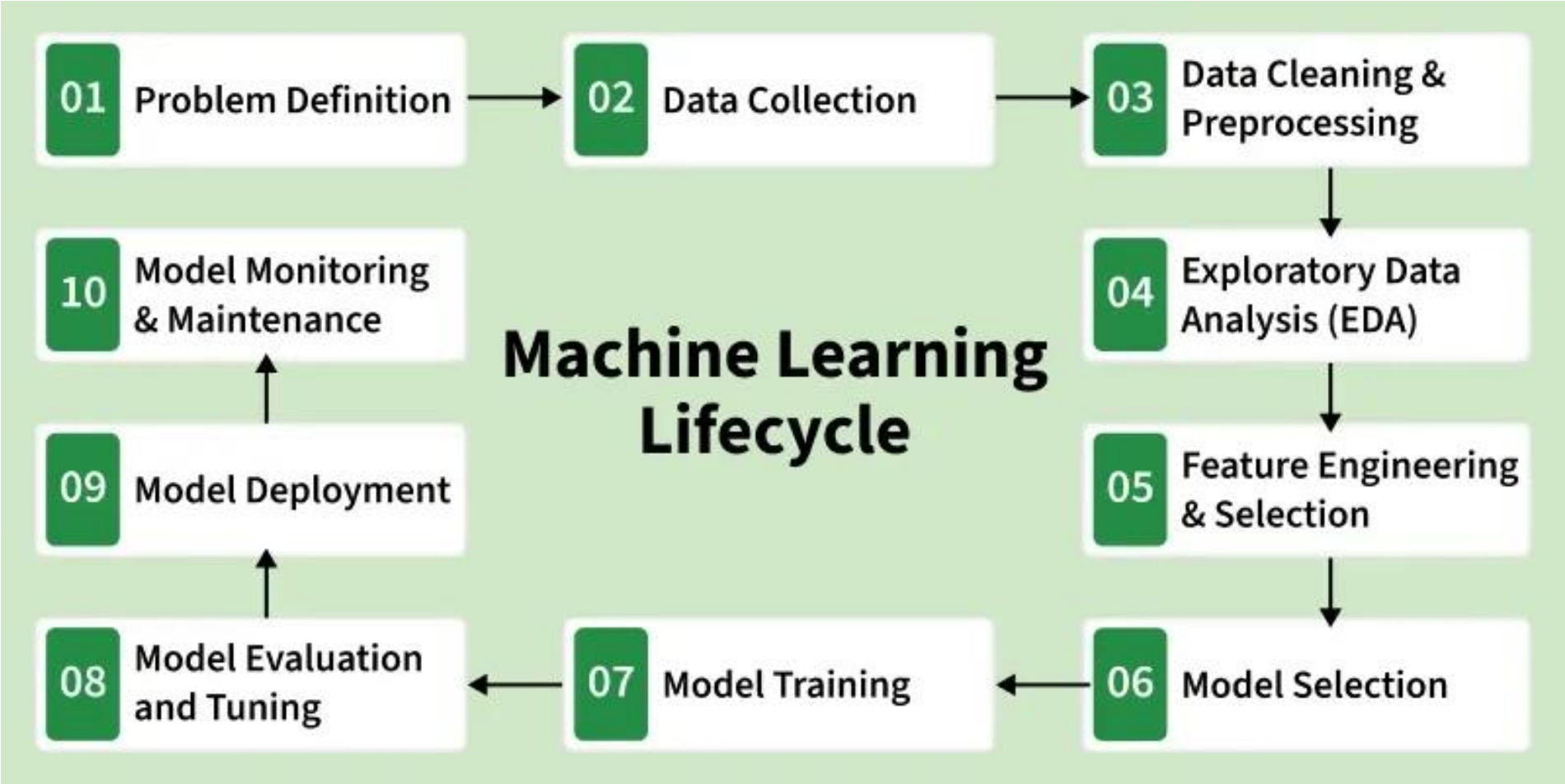
- Data Issues

1) Insufficient Quantity of Training Data
2) Non-representative Training Data
3) Poor-Quality Data
4) Irrelevant Features
5) Data Mismatch

- Algorithms

1) Overfitting the Training Data
2) Underfitting the Training Data

Machine Learning Cycle



Prediction Types : Classification vs. Numeric Prediction

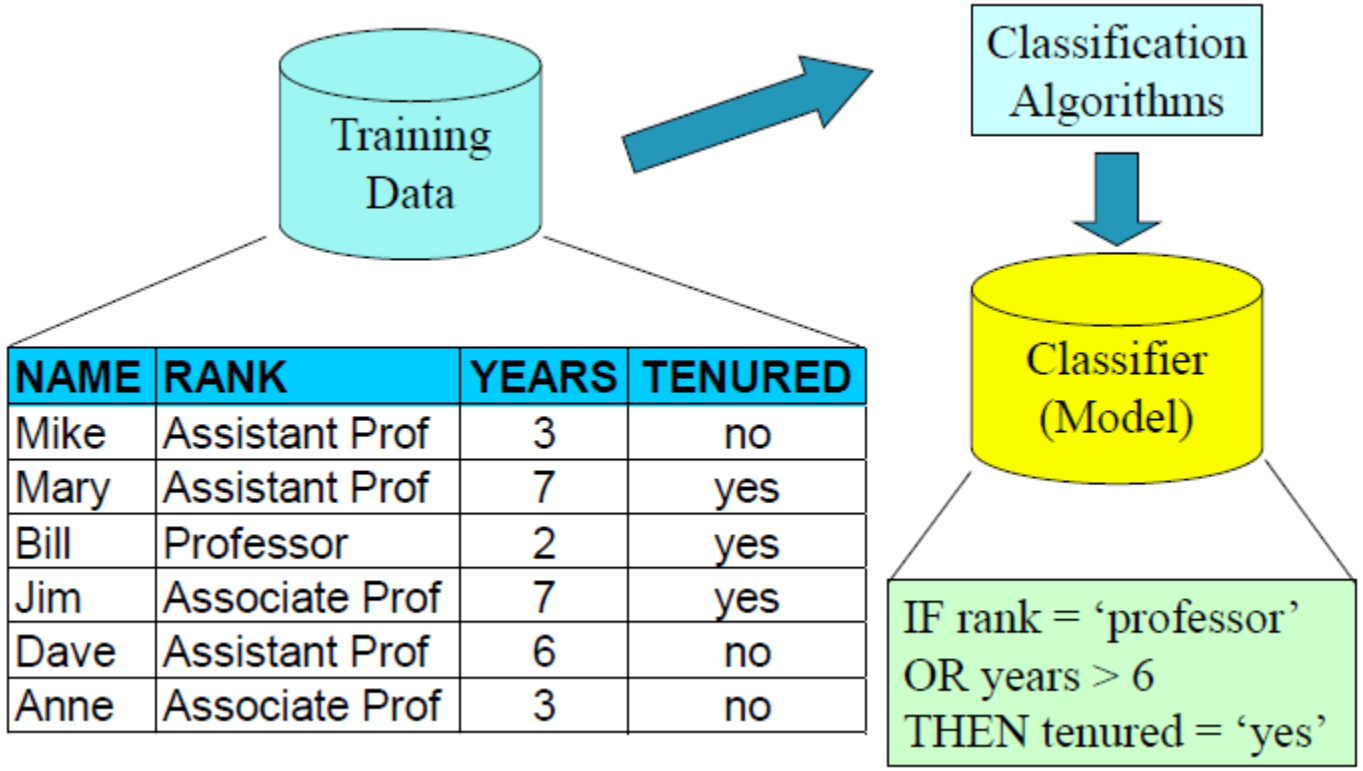
- Classification
  - Predicts categorical class labels (discrete or nominal)
  - Classifies data based on the training set (constructs a model) and the values (class labels) in a classifying attribute and uses it in classifying new data
- Numeric Prediction
  - Models continuous valued functions, i.e., predicts unknown or missing values

Classification—A Two-Step Process

1) Model construction : describing a set of pre determined classes
   - Each tuple/sample is assumed to belong to a predefined class, as determined by the class label attribute
   - The set of tuples used for model construction is training set
   - The model is represented as classification rules, decision trees, or mathematical formulae
2) Model usage : for classifying future or unknown objects
   - Estimate accuracy of the model
     - The known label of test sample is compared with the classified result from the model
     - Accuracy rate is the percentage of test set samples that are correctly classified by the model
     - Test set is independent of training set (otherwise overfitting)
   - If the accuracy is acceptable, use the model to classify new data

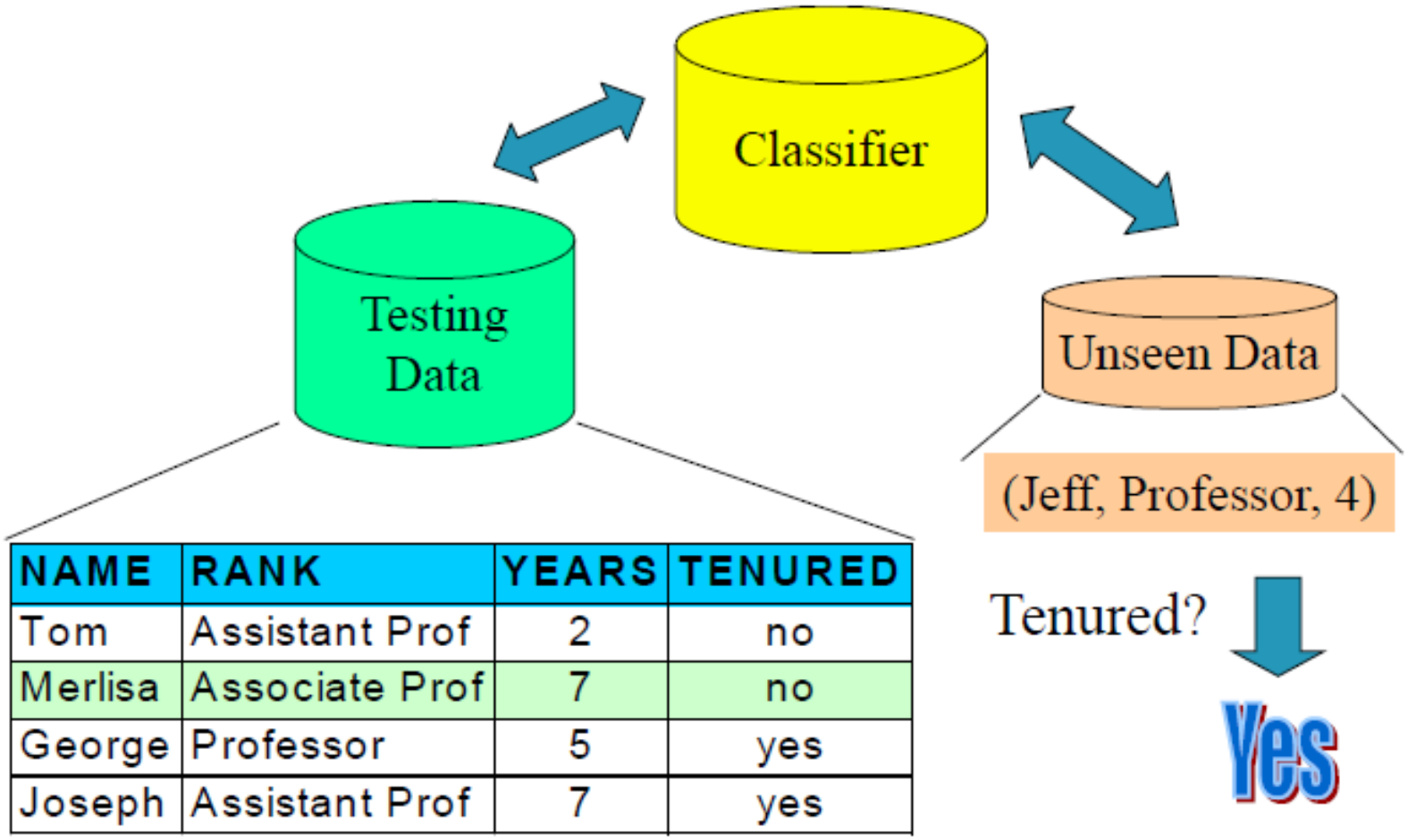
- Note: If the test set is used to select models, it is called validation set

Process (1): Model Construction



NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

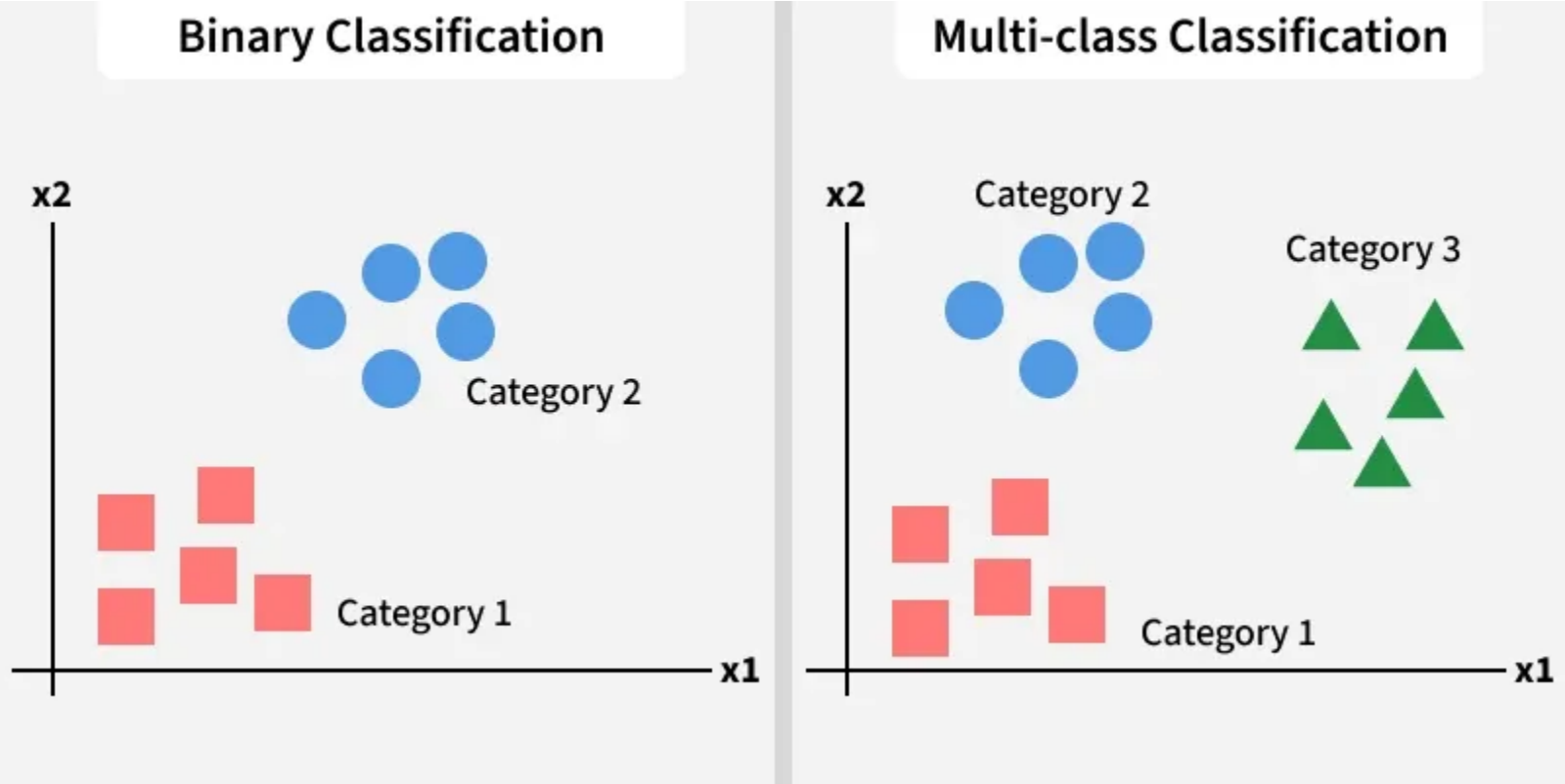
Process (2): Using the Model in Prediction



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

Types of Classification

1. Binary Classification (Two options)
2. Multiclass Classification (More Than Two)



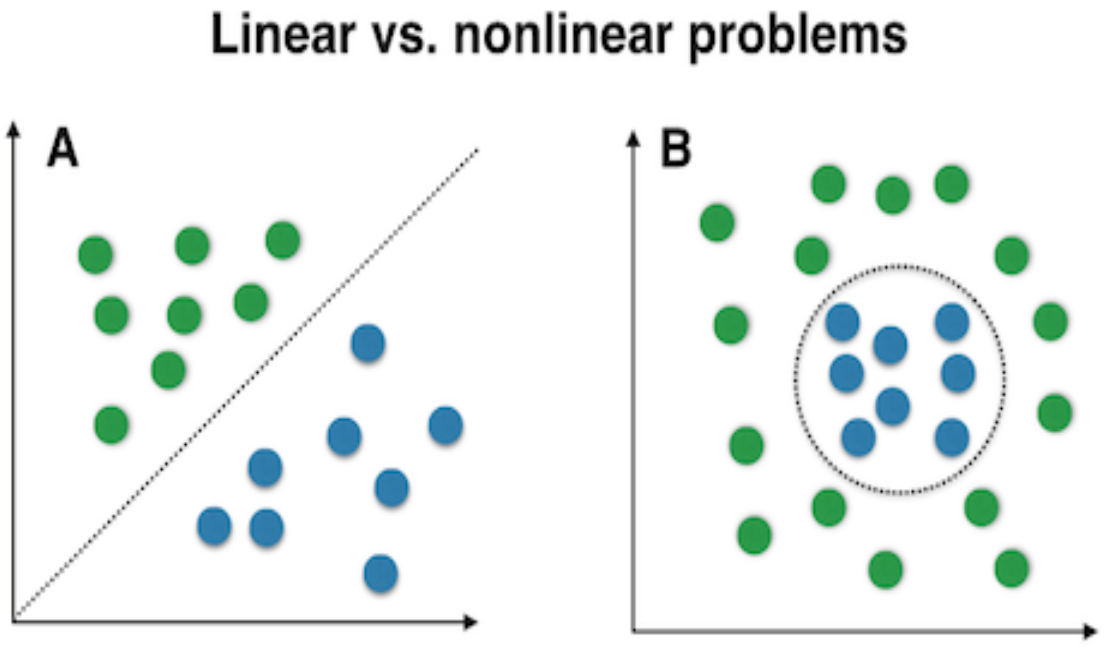
Binary Classification

Can a Line Separate Classes?

y=f(x), where y is the categorical output

Types of ML classification algorithms :

1. Linear models:
   - Logistic Regression
   - Support Vector Machine
2. Non-Linear Models
   - K-Nearest Neighbors
   - Decision Tree
   - Random Forest
   - Naive Bias



Linear Models

We have data like:

Hours Studied	Passed Exam
2	0
5	1
3	0
8	1

We want a rule that says:
Given hours studied → predict 0 or 1.

Linear Models

$$z=\omega*x+b$$

Prediction score = (weight × input) + bias

If :

- ω is positive → more hours increases score
- ω is negative → more hours decreases score

Linear Models

- Example:

If $w = 1$ and $b = -4$ Then:

- $x = 2 \rightarrow z = -2$ So low hours → negative score
- $x = 8 \rightarrow z = 4$ High hours → positive score

Already this separates classes.

If $z > 0 \rightarrow$ predict 1
If $z < 0 \rightarrow$ predict 0

That's a classifier.

The boundary is where: $z = 0$

Which means: $\omega x + b = 0$

That's just the dividing point.

Probability

Problem:

The model says:

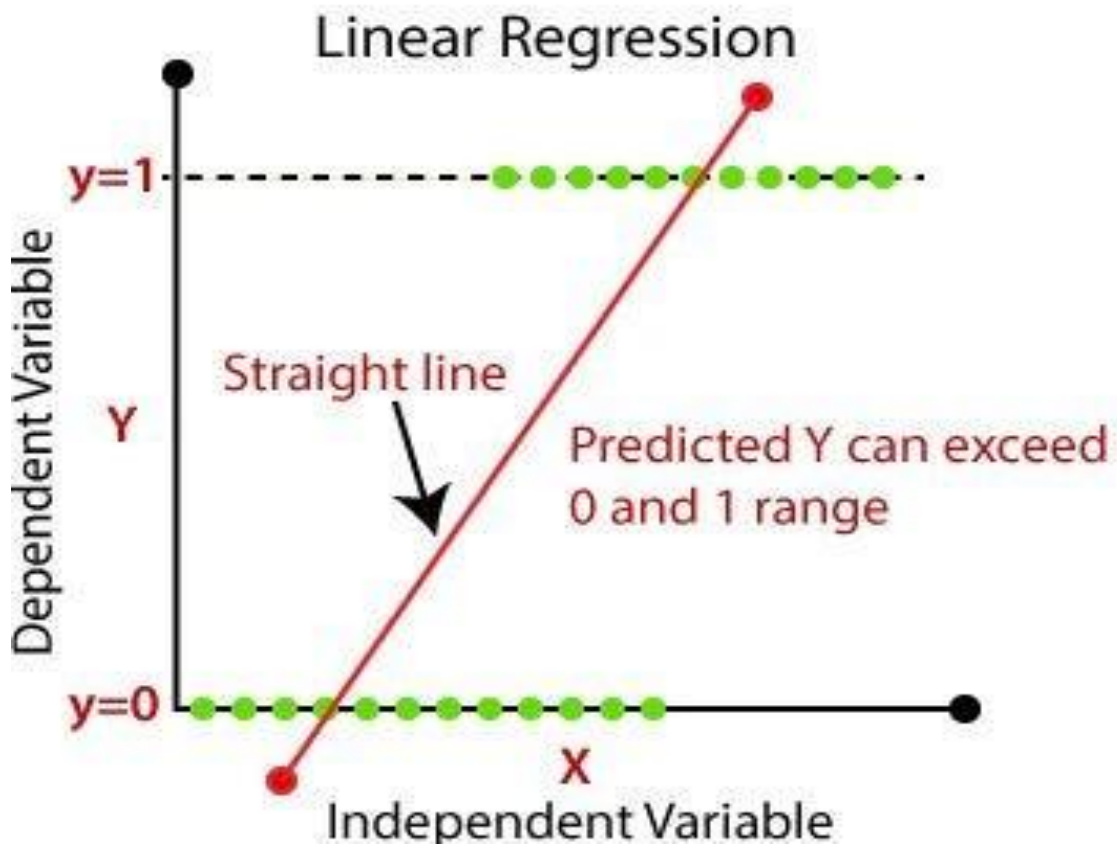
$x = 4.1 \rightarrow$ class 1

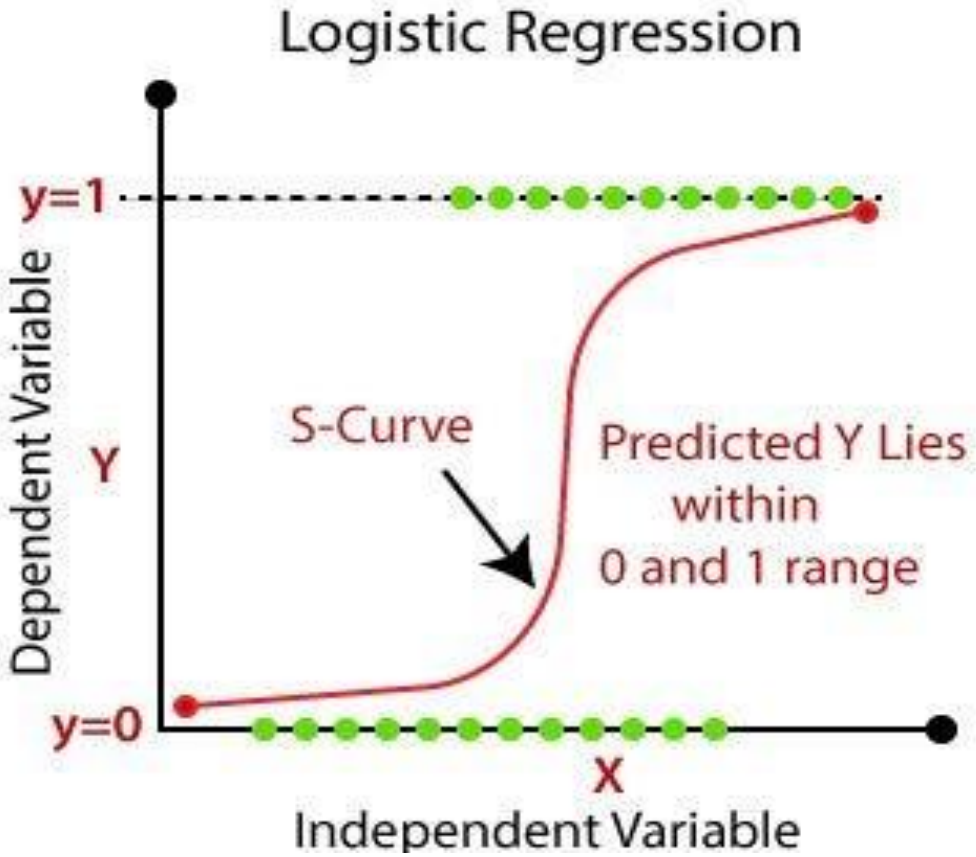
$x = 3.9 \rightarrow$ class 0 But they are almost identical.

Instead of hard 0 or 1, we want: A number between 0 and 1.

- 0.9 → very confident pass
- 0.6 → maybe pass
- 0.2 → likely fail

Logistic Regression





Interpretation: maps (-∞, ∞) to (0,1)

Logistic Regression

$$\sigma(z) = \frac{1}{1+e^{-z}} \text{ , } z = \omega * x + b$$

How Do We Make It Learn? (How do we choose ω and b?)

We introduce penalty when prediction is wrong.

- Case 1:

True label = 1

- Model says probability = 0.9 → good
- Model says probability = 0.1 → very bad

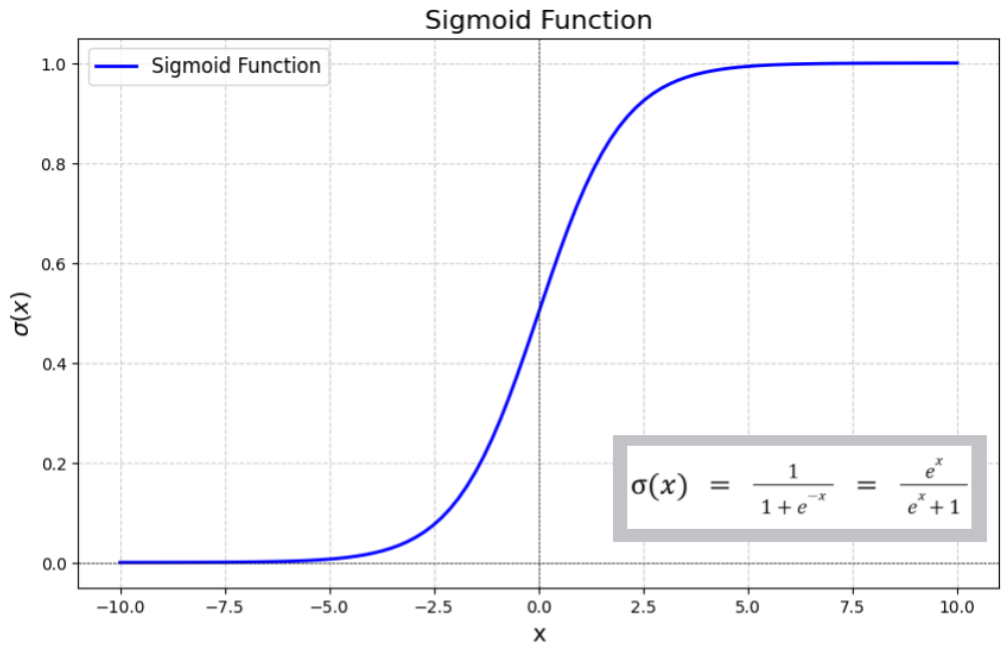
- Case 2:

True label = 0

- Model says probability = 0.1 → good
- Model says probability = 0.9 → very bad

We want a function that:

Punishes confident wrong predictions strongly.



Cross-Entropy

That function is:

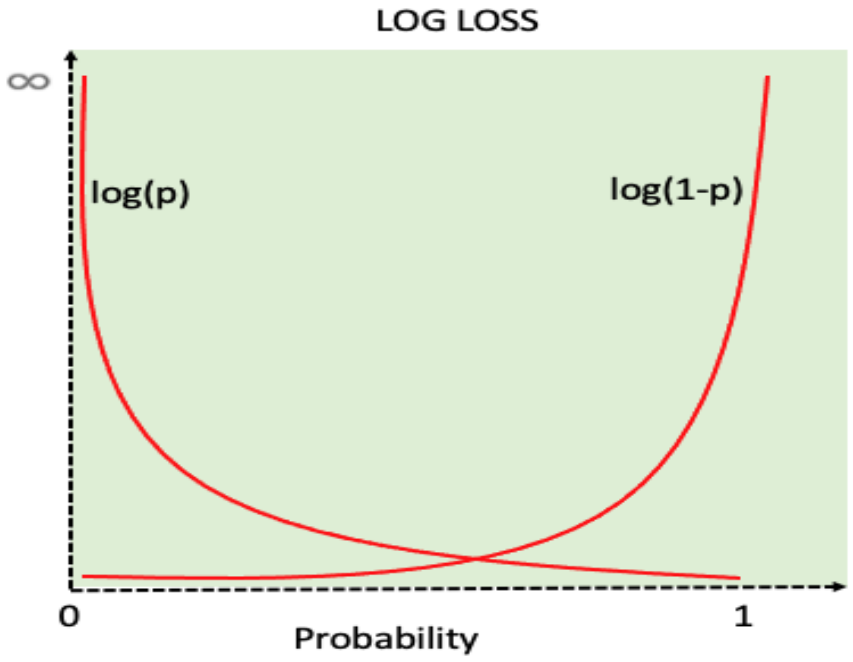
$$H(P,Q) = -\sum_i y_i * log(p_i)$$

$$Loss = -(y * \log(p) + (1-y) * \log(1-p))$$

- Where

y = true label (0 or 1)

p = predicted probability



We want a loss that:

- is small when prediction is correct
- becomes very large when we are confidently wrong

Example:

Two classes [Cat, Dog] : Dog is correct

$$Q = [0,1]$$

Prediction:

$$P = [0.9,0.1]$$

Cross entropy becomes:

$$H = -(0*\log(0.9)+1*\log(0.1))$$

$$H = -\log(0.1) = 2.302$$

If prediction was correct:

$$P = [0.1,0.9]$$

$$H = -\log(0.9) = 0.105$$ Much smaller.

Gradient Descent

- Our loss for one data point is:

$$L = -[\mathrm{y} * \log(p) + (1 - y) * \log(1 - p)] =$$

Where:

$$p = \sigma(z)$$

$$z = \omega * \mathrm{x} + \mathrm{b}$$

- So there is a chain:

$$\omega \rightarrow z \rightarrow \mathrm{p} \rightarrow \mathrm{L}$$

If we change ω a little bit, how does the loss change?

Gradient Descent

- Mathematically:

$$\frac{dL}{d\omega}$$

- So we follow the chain.

Because w affects z.
z affects p.
p affects L.

- So we use the chain rule:

$$\frac{dL}{d\omega} = \frac{dL}{dp} \cdot \frac{dp}{dz} \cdot \frac{dz}{d\omega}$$

Gradient Descent

$$L = -[\mathrm{y} * \log(p) + (1 - y) * \log(1 - p)]$$

Derivative with respect to p:

$$\frac{dL}{dp} = -(\frac{y}{p} - \frac{1-y}{1-p})$$

Derivative with respect to z:

$$\mathrm{p}=\sigma(\mathrm{z})=\frac{1}{1+\mathrm{e}^{-\mathrm{z}}}$$

$$\frac{dp}{dz}=\mathrm{p}*(1\text{-}\mathrm{p})$$

Derivative with respect to ω:

$$\frac{dz}{d\omega} = x$$

Multiply Everything

$$\frac{dL}{d\omega} = -\left(\frac{y}{p} - \frac{1-y}{1-p}\right).\mathrm{p}.(1-\mathrm{p}).\mathrm{x}$$

After simplifying

$$\frac{dL}{d\omega} = (p-y).x$$

- Thus the gradient is:

(Prediction − Truth) times input

- If:

  Prediction too large → gradient positive → weight decreases
  Prediction too small → gradient negative → weight increases

- So learning becomes:

$$w := w - \eta(p-y)x := w - \eta\frac{dL}{d\omega}$$

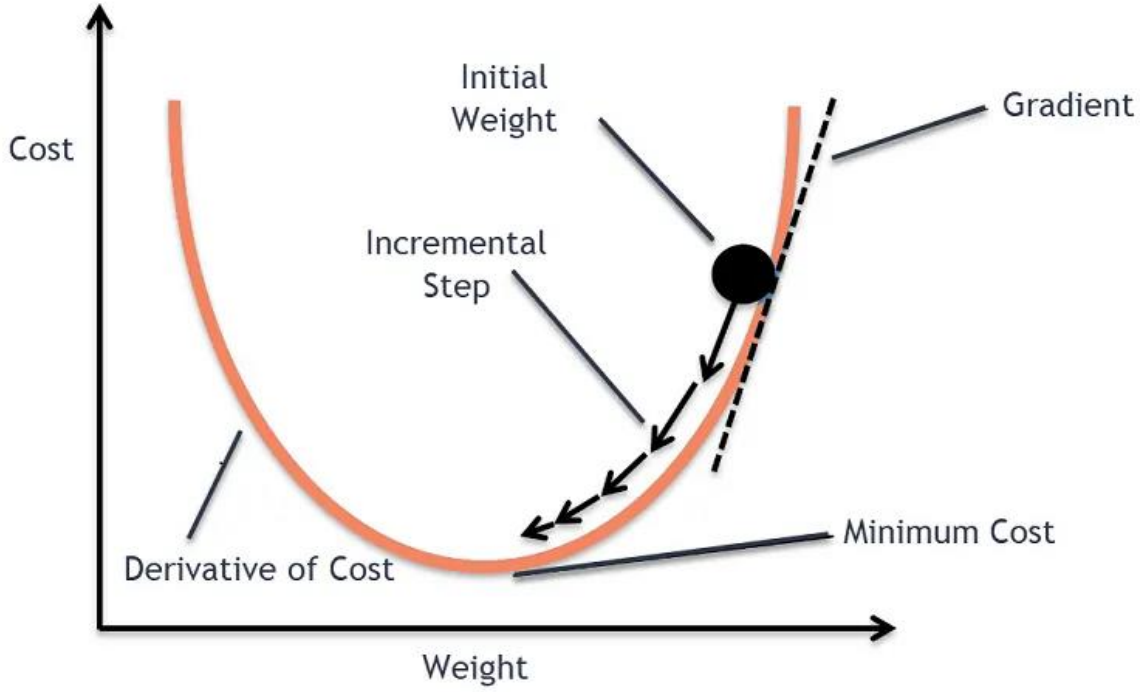
η: Learning Rate

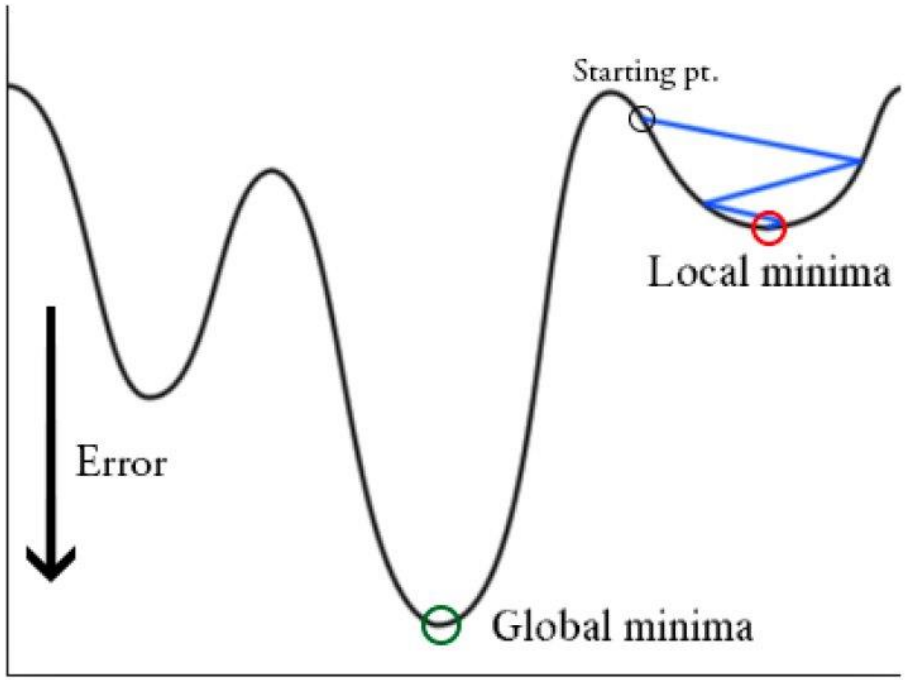
$$New\ weight = Old\ weight - \text{Learning Rate} * [(prediction - truth) * \text{input}]$$

Same for bias b

$$b := b - \eta\frac{dL}{db}$$

Gradient Descent





How Does This Work for the Whole Dataset?

We have N samples then Loss over dataset:

$$J = \frac{1}{N}\sum_{i=1}^{N} L_i$$

Gradient becomes:

$$\frac{dJ}{dw} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)x_i$$

Bias:

$$\frac{dJ}{db} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)$$

Multiple Features

Instead of one feature x, we have:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Weights:

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_d \end{bmatrix}$$

Linear model:

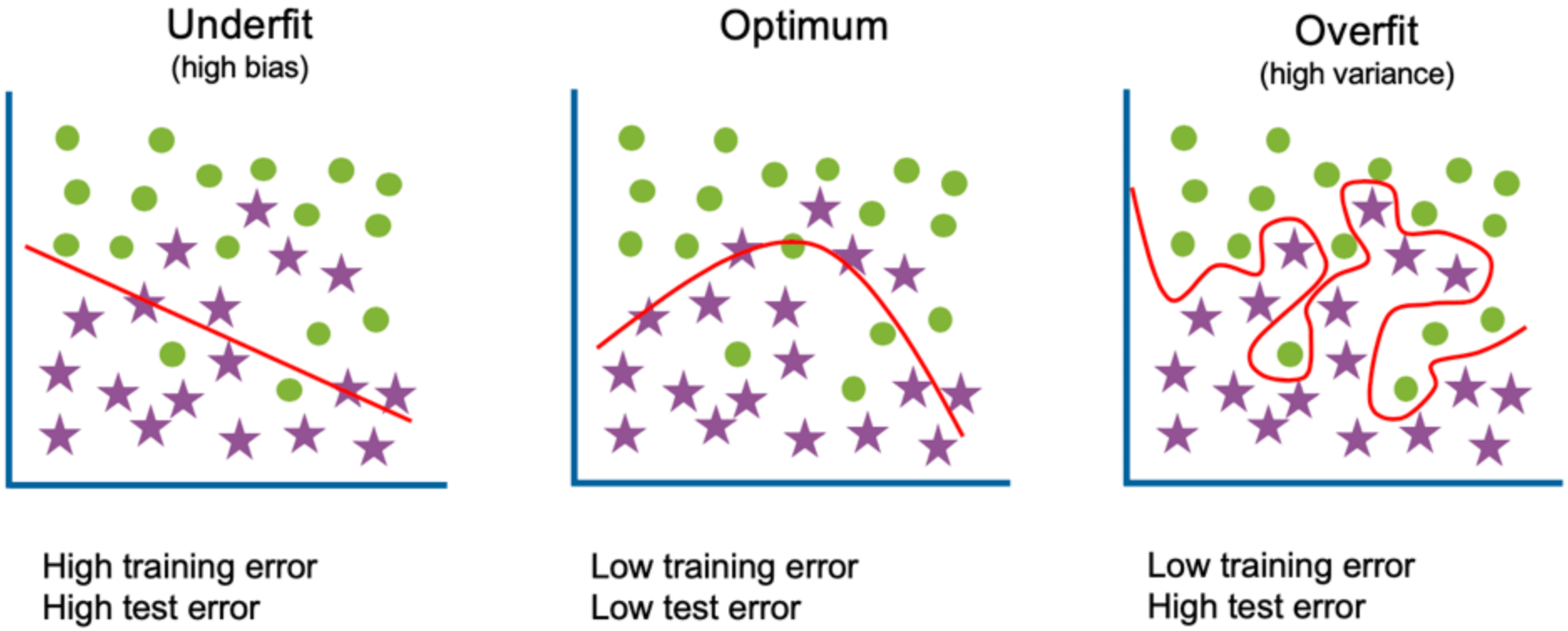
$$z = \omega^t x + b$$

Which expands to:

$$\omega_1 x_1 + \omega_2 x_2 + \cdots + \omega_d x_d + b$$

Underfitting vs Overfitting

- If our model performs perfectly on training data, is that enough?



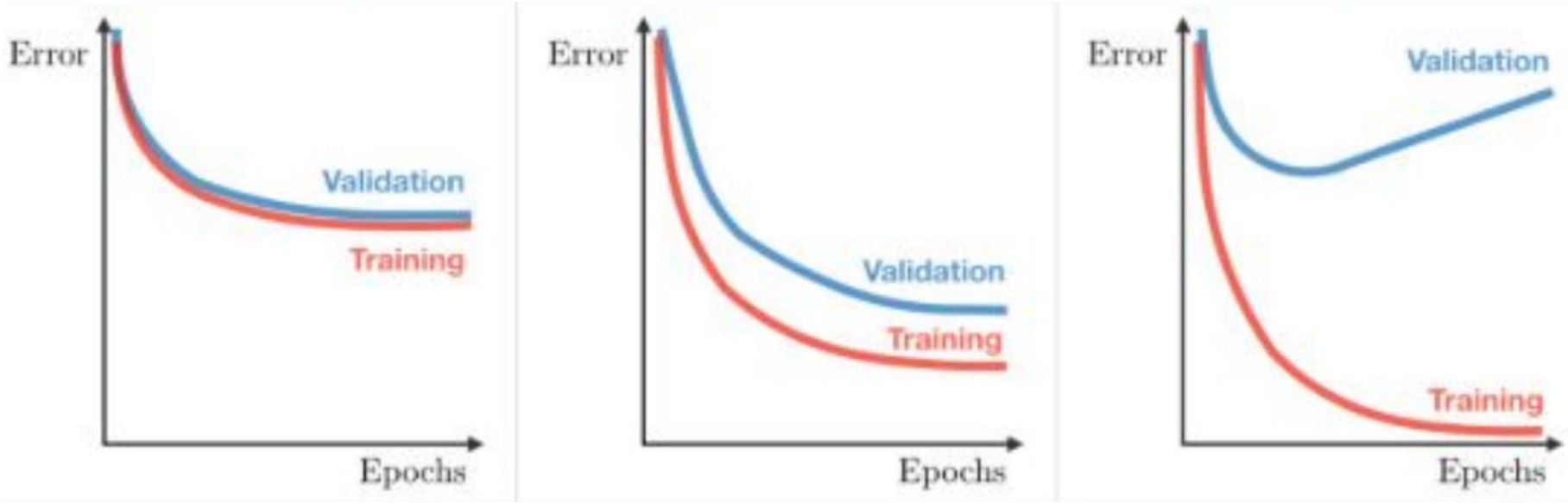
Straight line trying to fit curved data.

Crazy wiggly boundary fitting every point.
Small dataset

Underfitting vs Overfitting

- Underfitting
  - The model is too simple to capture the true pattern.
  - Not enough training time
  - Too few features

- Overfitting
  - The model memorizes the training data instead of learning the pattern.
    - Poor performance on new (test) data
    - Model is too complex
    - Too many features
    - Not enough training data



Bias and Variance

Bias

Error due to model assumptions.

In linear models:

Assuming data is linearly separable when it is not.

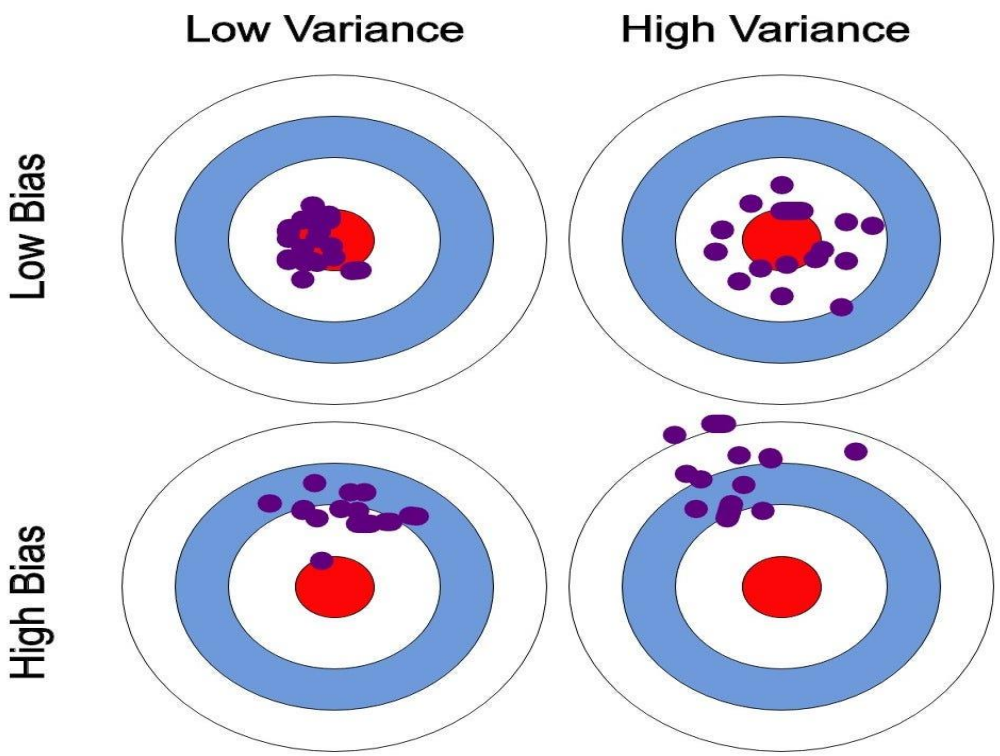
High bias → underfitting.

Variance

Error due to sensitivity to data fluctuations.

If dataset is changed slightly and model changes drastically → high variance.

High variance → overfitting



Addressing overfitting

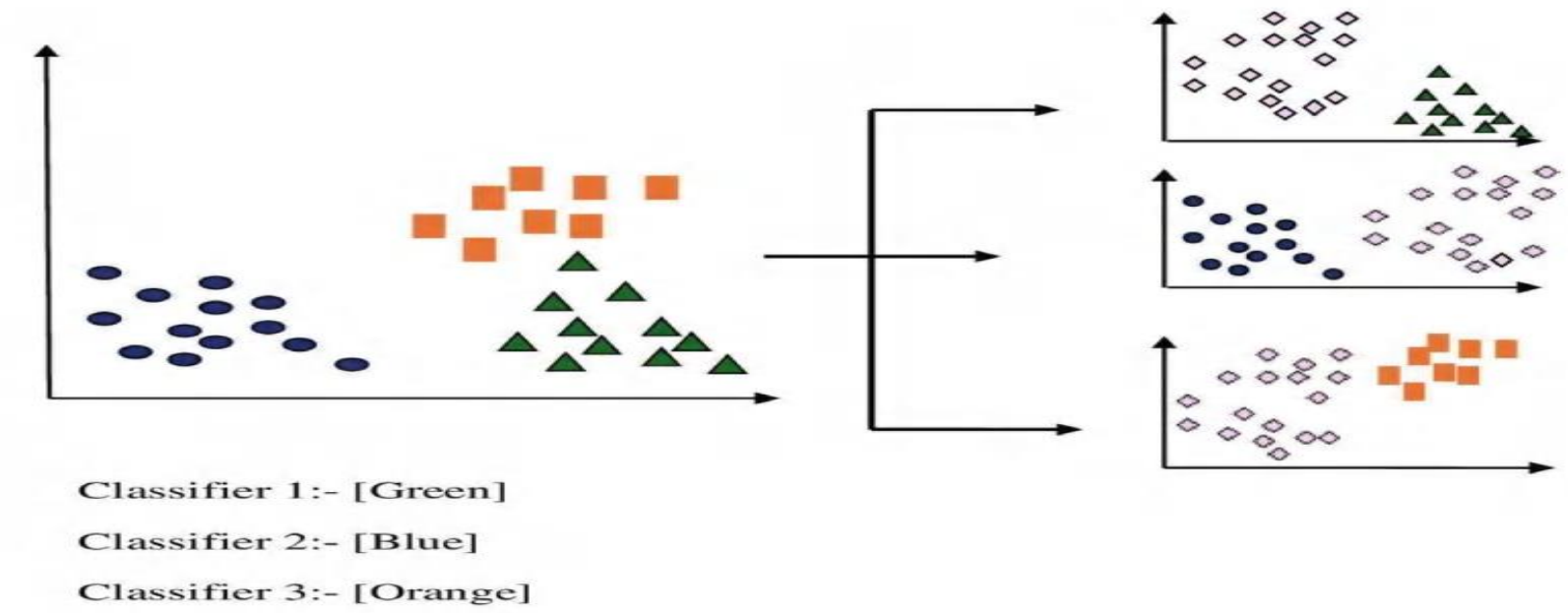
1. Collect more data.
2. Reduce number of features.
   — Manually select which features to keep.
   — Model selection algorithm
3. Regularization.
   — Keep all the features but reduce magnitude/values of parameters .
   — Works well when we have a lot of features, each of which contributes a bit to predicting

$$J = CrossEntropy + \lambda \mid\mid w \mid\mid^2$$

$$\lambda \mid\mid w \mid\mid^2 = \lambda \sum_j \omega_j{}^2$$

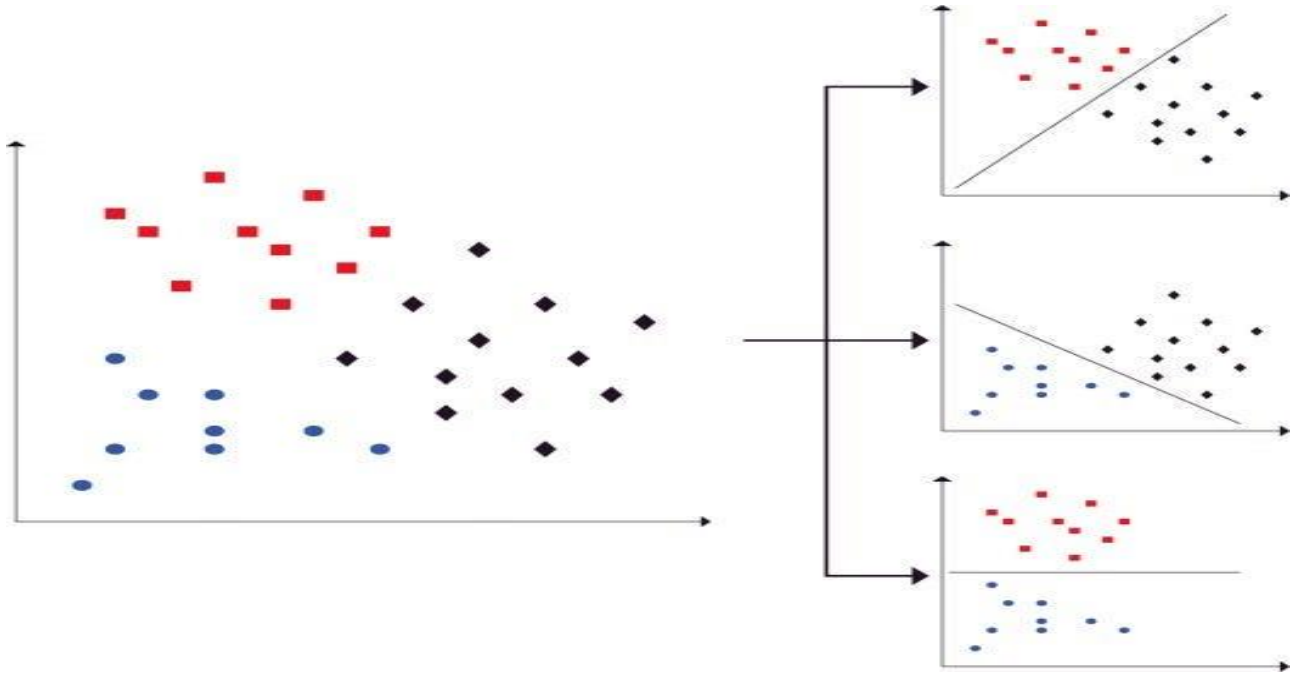
Binary Classifiers for Multi-Class Classification

- Binary classification models do not support multi-class classification
- One-vs-Rest strategy
  - Splits a multi-class classification into one binary classification problem per class.
  - One model to be created for each class.



Binary Classifiers for Multi-Class Classification

- One-vs-One strategy
  - Splits a multi-class classification into one binary classification problem per each pair of classes.
  - Splits the dataset into one dataset for each class versus every other class.
  - More datasets
  - **- N-class** instances dataset, $\frac{\boldsymbol{N*(N-1)}}{\mathbf{2}}$ binary classifier models.



http://vision.stanford.edu/teaching/cs231n-demos/linear-classify/

K-Nearest Neighbors

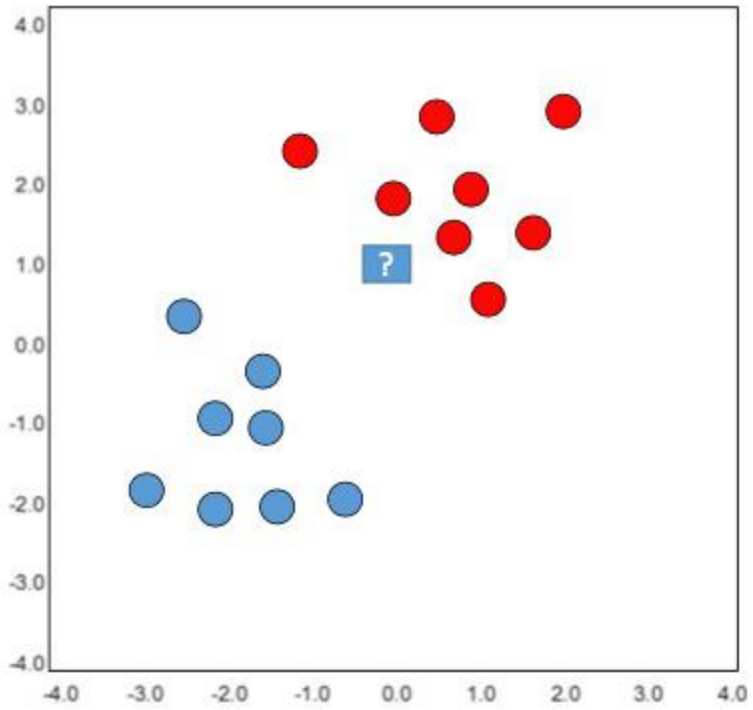
- K-Nearest Neighbors (KNN)
- How it Works?
- Distance Metrics
- K-NN Computational complexity

K-Nearest Neighbors (KNN)

- Instance-based learning:
  – Store training examples and delay the processing (“lazy evaluation”) until a new instance must be classified
- Two classes, two features

We want to assign label to unknown data point?

- Label should be red.



K-Nearest Neighbors (KNN)

Idea:

- We have similar labels for similar features.
- We classify new test point using similar training data points.

Algorithm overview:

- Given some new test point **x** for which we need to predict the class **y**.
- Find **most similar** data-points in the training data.
- Classify x "**like**" these most similar data points.

Questions:

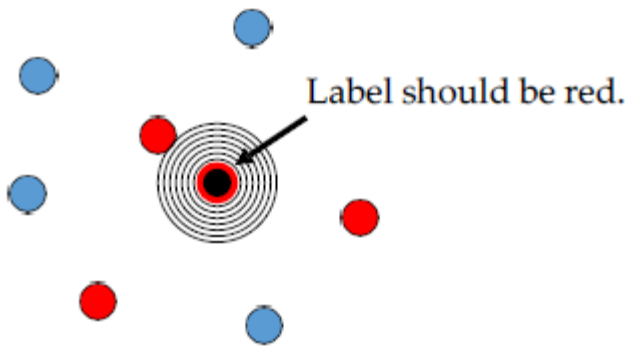
- How do we determine the similarity?
- How many similar training data points to consider?
- How to resolve inconsistencies among the training data points?

K-Nearest Neighbors (KNN)

❖ 1-Nearest Neighbor:

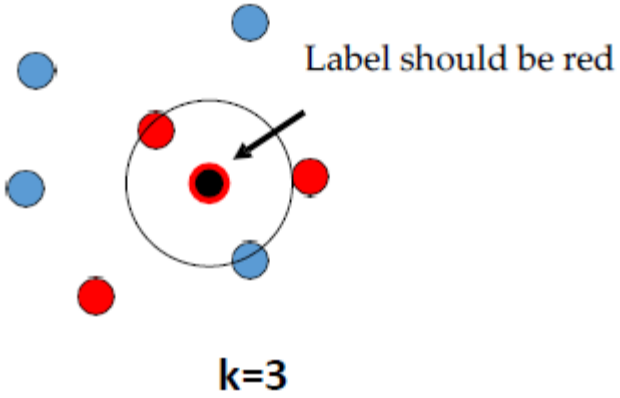
Simplest ML Classifier

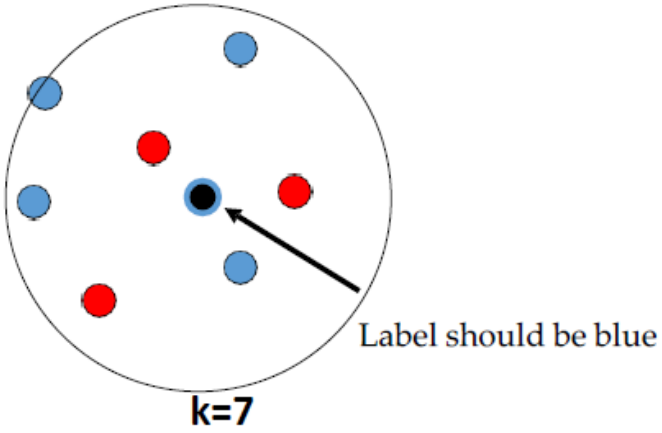
Idea: Use the label of the closest known point.



❖ Generalization:

Determine the label of k nearest neighbors and assign the most frequent label





Formal Definition:

- Using the S_x , we can define a classifier as a function that gives us the most frequent label of the data points in S_x

$$h(x) = mode(\{y' : (x', y') \in S_x\})$$

- $h(x)$ means the predicted label (output) for the input x.
- S_x represents the set of the K nearest training points to x.
- Each element in the dataset is written as: (x', y')
  - x' = the features of a training example
  - y' = the label of that training example
- $\{y' : (x', y') \in S_x\}$ ⇒ "Take the labels y'y' of all points whose pairs (x', y') are in the neighbor set Sx."

How it Works?

A. Selecting appropriate K which is number of nearest neighbors
   - Higher values of K reduces the noisy data and make the predictions more stable.
   - If k is too large the model may become too simple and miss important patterns (Underfitting)

Use for selecting K:

1. Cross-Validation
2. Elbow Method
3. Odd Values for k

Distance Metrics

B. Calculating the distance of new data point,

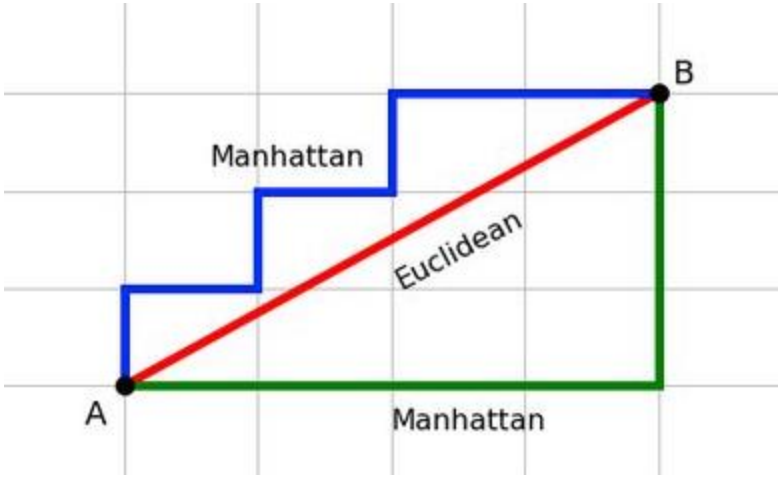
To identify nearest neighbor we use below distance metrics:

1. Euclidean Distance

$$d(x, y) = \sqrt{\sum_{j=1}^{d} (x_j - y_j)^2}$$

2. Manhattan Distance

$$d(x, y) = \sum_{i=1}^{n} | x_i - y_i |$$



Distance Metrics

3. **Minkowski Distance**

$$\mathrm{d}(\mathrm{x},\mathrm{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{\frac{1}{p}}$$

Where:

x_i, y_i = feature values

n = number of features (dimensions)

p = parameter that controls the type of distance

p value	Distance Type
p=1	Manhattan Distance
p=2	Euclidean Distance
p→∞	Chebyshev Distance

Distance Metrics

4. Cosine Distance

$$\text{cosine distance} = 1 - \text{cosine similarity}$$

$$cosine\ similarity(x, y) = \frac{x \cdot y}{||x||\ ||y||}$$

Where:

- x·y = dot product
- $||x||$,||y|| = vector magnitudes

Values range from:

1 → vectors point in the same direction

0 → vectors are orthogonal (unrelated)

-1 → opposite direction

Distance Metrics

Properties of Distance Metrics

- Nonnegative , dist(x,x') ≥ 0
- Symmetric , dist(x,x') = dist(x',x)
- dist(x,x') =0 $\Leftrightarrow x = x'$

- **Practical issues in computing distance:**
  - **Mismatch in the values of data**
    - **Issue:** Distance metric is mapping from d-dimensional space to a scaler. The values should be of the same order along each dimension.
  - **Solution:** Data Normalization

How it Works?

C. Select the K nearest neighbors based on distance calculated

D. Voting for Classification or Taking Average for Regression

Advantages:

- Can be used for both classification and regression
- Non parametric
- Works well on multi classes algorithms.
- unbiased in nature and also doesn’t make prior assumptions

- Disadvantages
  - Measuring the distance between all data points in training samples is a computationally expensive task
  - Requires more memory

K-NN Computational complexity:

- Training Complexity
  - Consists of **storing the dataset** → O(1)
- **Prediction**
  - Compute distance from the query point to all n samples → O(nd)
  - Find the k smallest distances → about O(n)
- **Total prediction complexity** →O(nd)
- **Memory**

The algorithm must store the dataset: → O(nd)